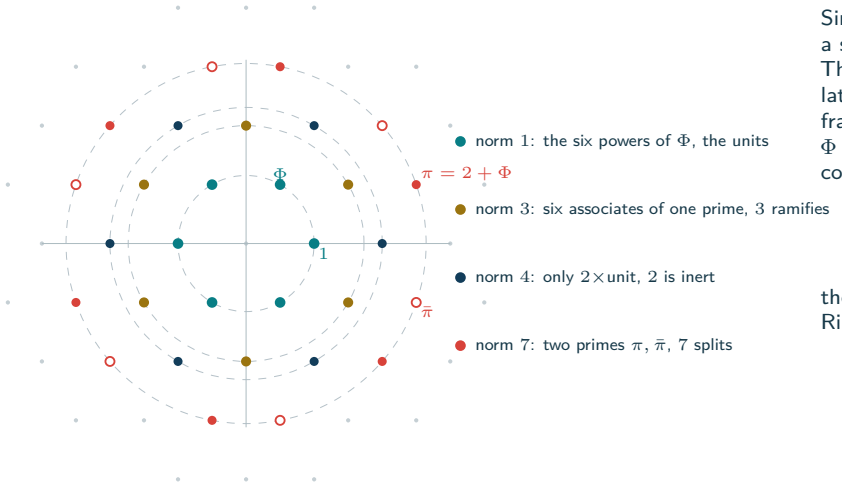
NUMBER THEORY ON ONE PAGE

The sixth root of unity: split, inert, ramified

The complex golden ratio $\Phi = e^{i\pi/3}$ generates the Eisenstein integers, where primes split, stay inert or ramify, and 3 ramifies.

1 / THE RULE AND THE PICTURE

Primes of $\mathbb{Z}$ factor in the lattice according to which norms occur



Since $\Phi = e^{i\pi/3}$ we have $\Phi^3 = -1$ and $\Phi^6 = 1$, so Φ is a sixth root of unity solving $x^2 - x + 1 = 0$.
The numbers $a + b\Phi$ with whole a, b form a hexagonal lattice, the ring $\mathbb{Z}[\Phi]$ (the Eisenstein integers), and with fractions a, b the field $\mathbb{Q}(\Phi) = \mathbb{Q}(\sqrt{-3})$. Using $\Phi + \bar{\Phi} = 1$ and $\Phi\bar{\Phi} = 1$ the norm, the number times its conjugate, is

$$N(a + b\Phi) = a^2 + ab + b^2$$

the squared distance from the origin in the lattice. Rings of equal norm show how a prime p of $\mathbb{Z}$ factors.

2 / THREE FATES FOR A PRIME

A prime p can split, stay inert or ramify

fate	what happens to p	which p	examples
split	$p = \pi\bar{\pi}$ with two different primes	$p \equiv 1 \pmod 3$	$7 = (2+\Phi)(2+\bar{\Phi})$, $13 = (3+\Phi)(3+\bar{\Phi})$
inert	p stays prime	$p \equiv 2 \pmod 3$	2, 5, 11: no element has norm p
ramified	$p =$ unit $\times\, \pi^2$	$p = 3$	$3 = \Phi\,(2-\Phi)^2$

The norm of Φ is 1, so Φ is a unit and carries no prime. The prime 3 is not prime in $\mathbb{Z}[\Phi]$ since $3 = N(2-\Phi) = (2-\Phi)(2-\bar{\Phi})$, and the two conjugate factors differ only by the unit Φ, which is why 3 ramifies instead of splitting.

3 / READING THE FATE FROM A POLYNOMIAL

Factor $x^2 - x + 1$ modulo p

p	$x^2 - x + 1$ modulo p	type	fate in $\mathbb{Z}[\Phi]$
2	$x^2 - x + 1$ has no root	irreducible	inert
3	$x^2 - x + 1 = (x+1)^2$	a repeated root	ramified
5	no root	irreducible	inert
7	$(x-3)(x-5)$	two different roots	split
11	no root	irreducible	inert
13	$(x-4)(x-10)$	two different roots	split

Because $\mathbb{Z}[\Phi] = \mathbb{Z}[x]/(x^2 - x + 1)$, the polynomial modulo p factors just as p does: two different roots mean split, no root means inert, and a repeated root means ramified.

4 / THE RAMIFIED PRIME HAS A NAME, AND THE REAL GOLDEN FIELD BEHAVES ALIKE

$\Phi - \bar{\Phi} = \sqrt{-3}$, and $\varphi - \bar{\varphi} = \sqrt{5}$

	complex golden: $\mathbb{Z}[\Phi]$	real golden: $\mathbb{Z}[\varphi]$
polynomial	$x^2 - x + 1$	$x^2 - x - 1$
norm of $a + b\tau$	$a^2 + ab + b^2$	$a^2 + ab - b^2$
units	six, the powers of Φ	infinitely many, $\pm\varphi^n$
ramified prime	3, and $3 = -(\sqrt{-3})^2$	5, and $5 = (\sqrt{5})^2$
splits when	$p \equiv 1 \pmod 3$	$p \equiv \pm 1 \pmod 5$
inert when	$p \equiv 2 \pmod 3$	$p \equiv \pm 2 \pmod 5$

The difference of the two roots of $x^2 - x \pm 1$ generates the ramified prime: $\Phi - \bar{\Phi} = 2\Phi - 1 = \sqrt{-3}$ and $\varphi + \frac{1}{\varphi} = 2\varphi - 1 = \sqrt{5}$. In the real field the fate depends on $p \bmod 5$, in the complex field on $p \bmod 3$.

5 / WHY IT DEPENDS ONLY ON $p \bmod 3$

The sixth roots of unity are the cube roots of unity with a sign

Since $\mathbb{Q}(\zeta_6) = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\Phi)$, the cyclotomic rule applies: if f is the order of p modulo 3 then p splits into $2/f$ primes. For $p \equiv 1$ the order is $f = 1$ and p splits into two, for $p \equiv 2$ the order is $f = 2$ and p stays inert, and $p = 3$ divides the conductor and ramifies.

ASIDE / TRY THESE

Is 17 prime in $\mathbb{Z}[\Phi]$? $17 \equiv 2 \pmod 3$, so yes. Does 19 split? $19 \equiv 1$, and $19 = 2^2 + 2 \cdot 3 + 3^2 = N(2 + 3\Phi)$.

LINKS TO OTHER SHEETS

The golden-type numbers and cyclotomic sheets; the adding-a-new-number sheet gets a new case, 3 ramifies with $\sqrt{-3}$.

Convention. $N(a + b\Phi) = a^2 + ab + b^2$; $\pi = 2 + \Phi$; fates checked from $x^2 - x + 1$ modulo p for $p \le 19$, and from the lattice counts 6, 6, 6, 12 at norms 1, 3, 4, 7.

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